

BINF 4002

Lecture 3: Probability

Today's Class: Probability

Outline

1. What are random variables?
2. Expectation
3. Joint random variables

A Probability Triple is samples, events, and a measure

Formally, a probability triple (X, Σ, P) is a tuple of three things:

1. A set of possible samples the random variable can take: X

A Probability Triple is samples, events, and a measure

Formally, a probability triple (X, Σ, P) is a tuple of three things:

1. A set of possible samples the random variable can take: X
2. The set of possible “events” that you are allowed to consider over these samples: $\Sigma \subseteq 2^X$

A Probability Triple is samples, events, and a measure

Formally, a probability triple (X, Σ, P) is a tuple of three things:

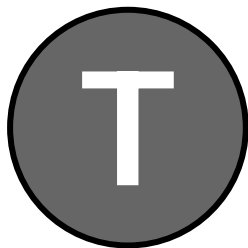
1. A set of possible samples the random variable can take: X
2. The set of possible “events” that you are allowed to consider over these samples: $\Sigma \subseteq 2^X$
3. A measure over these samples and events that is a valid probability measure:
 $P: \Sigma \rightarrow [0, 1]$.
 - a. $P(X) = 1$
 - b. For disjoint $A_i \in \Sigma$, $P(\cup_i A_i) = \sum_i P(A_i)$

Discrete probability triples: Coin flips!

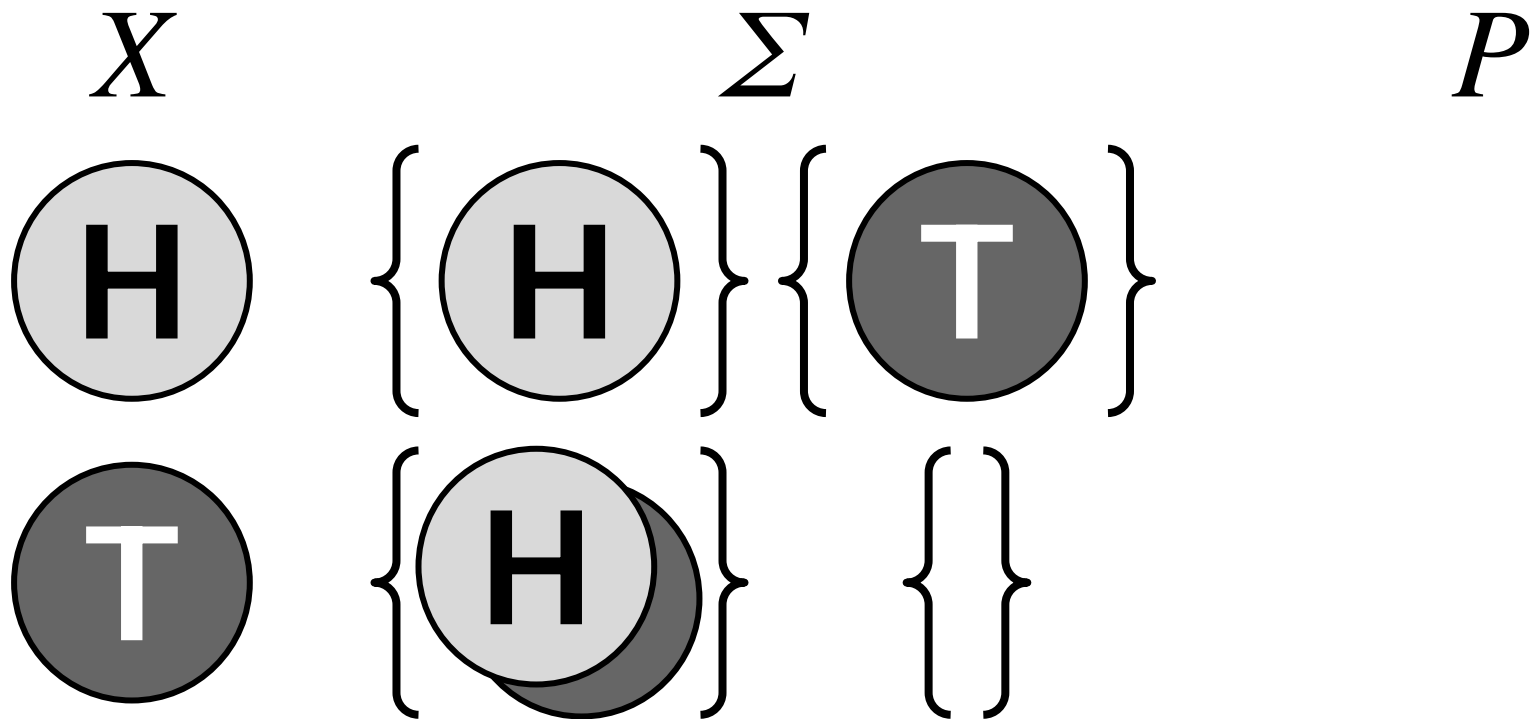
X

Σ

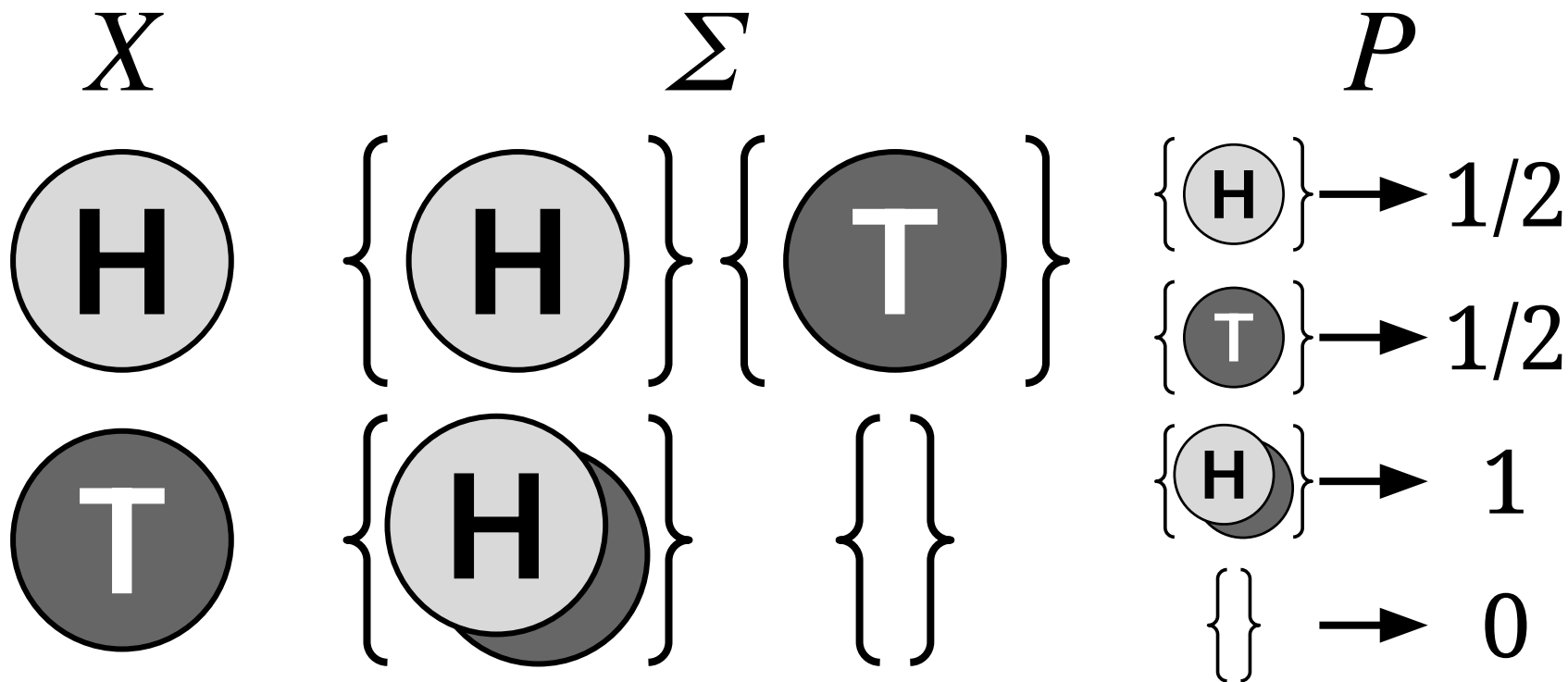
P



Discrete probability triples: Coin flips!



Discrete probability triples: Coin flips!



Random Variable

Formally, a random variable x is a *function* from the sample space X of a probability triple to a measurable space E of the observations of x .

Any probability triple defines a random variable under the identity function, and any random variable induces a probability triple over E ...

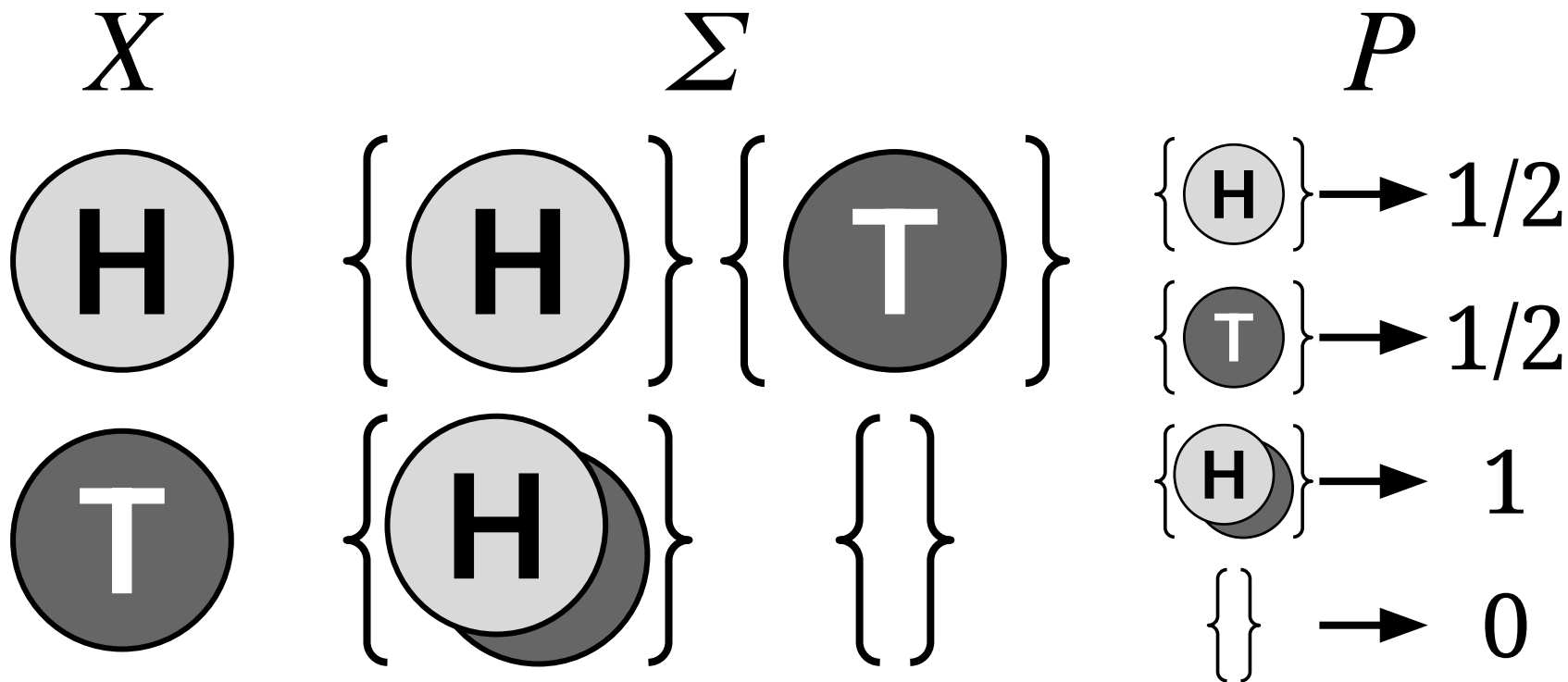
Any probability triple defines a random variable under the identity function, and any random variable induces a probability triple over E ...

So you can think of these two concepts as more or less equivalent.

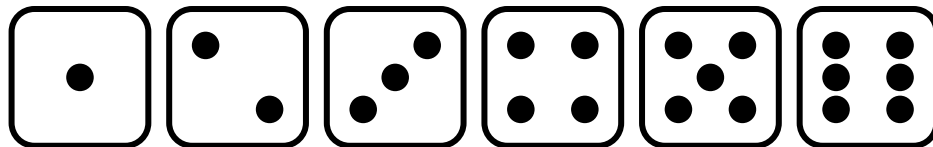
Warning!

Traditionally, these two things are not conflated.

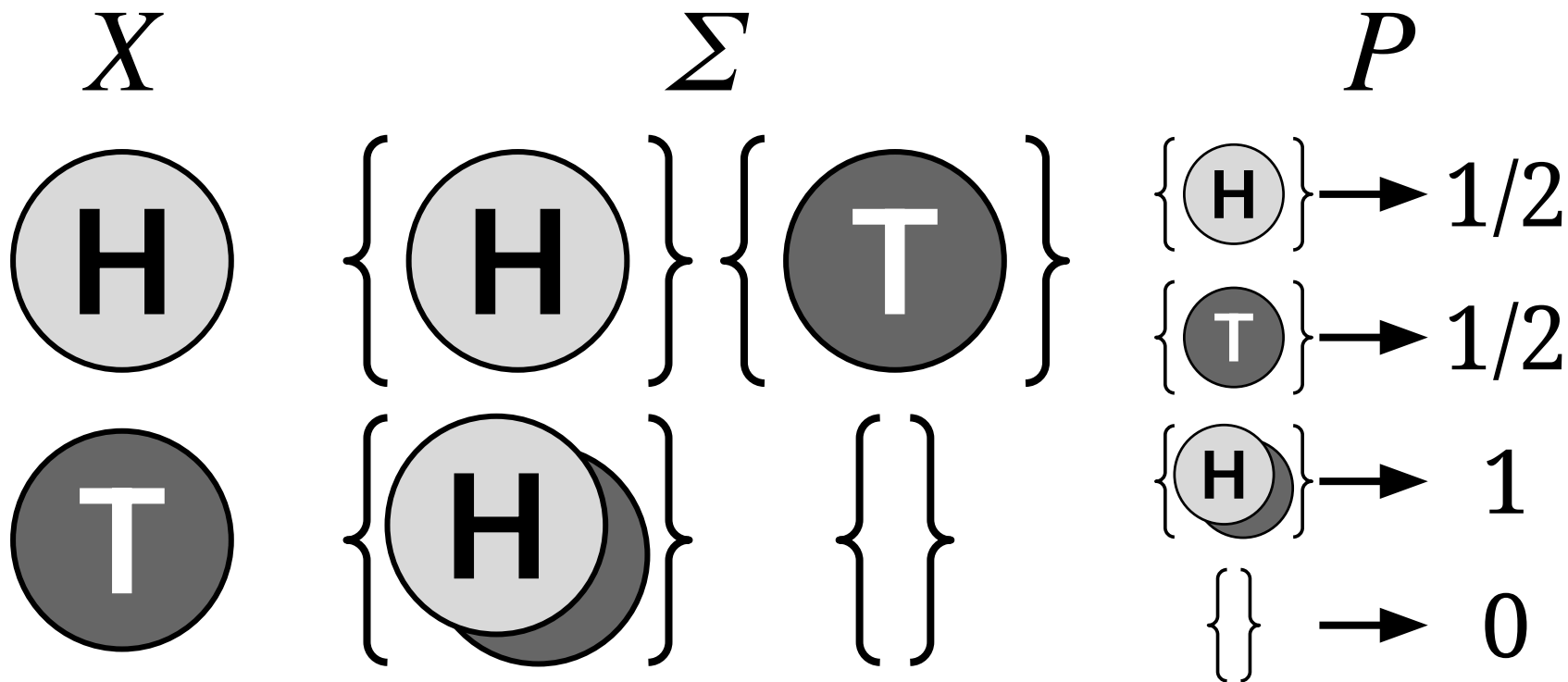
Discrete probability triples: Coin flips!



Discrete probability triples: Dice rolls!



Discrete probability triples: Coin flips!



Discrete probability triples: Coin flips!



P	
$\{H\}$	$\rightarrow 1/2$
$\{T\}$	$\rightarrow 1/2$
$\{H, T\}$	$\rightarrow 1$
$\{\}$	$\rightarrow 0$

P is defined by its value under elements of X !



P	
$\{H\}$	$\rightarrow 1/2$
$\{T\}$	$\rightarrow 1/2$
$\{H, T\}$	$\rightarrow 1$
$\{\}$	$\rightarrow 0$

Why? Because unions sum!

Formally, a probability triple (X, Σ, P) is a tuple of three things:

1. A set of possible samples the random variable can take: X
2. The set of possible “events” that you are allowed to consider over these samples: $\Sigma \subseteq 2^X$
3. A measure over these samples and events that is a valid probability measure:
 $P: \Sigma \rightarrow [0, 1]$.
 - a. $P(X) = 1$
 - b. For disjoint $A_i \in \Sigma$, $P(\cup_i A_i) = \sum_i P(A_i)$

Problem! What about
infinite X ?

P is defined by its value under “atoms” of X !



What is an “atom” for a set X ?

Intuitively

1. Atoms are the “smallest” parts of X .
2. Anything that we can measure in X must be made up of atoms.
3. Atoms all have the same “size” (not the same probability).
4. The probability of an event consisting of any set of things in X is the sum of the probability over the atoms making it up.

Informal use of atoms

- Suppose x is some “atom” of X .
- Let’s use the notation $dP(x)$ to refer to the probability dP of the atom x .
- Then, for an event $S \in \Sigma$, $\sum_S dP(x)$ is the sum of the probabilities of all the atoms of X in S (and thus is the “probability” of S).
- This means that $\sum_S dP(x) = P(S)$

Infinite sums represented using integrals $\int$

- Suppose x is some “atom” of X .
- Let’s use the notation $dP(x)$ to refer to the probability dP of the atom x .
- Then, for an event $S \in \Sigma$, $\int_S dP(x)$ is the sum of the probabilities of all the atoms of X in S (and thus is the “probability” of S).
- This means that $\int_S dP(x) = P(S)$

Infinite sums represented using integrals $\int$

- Suppose x is some “atom” of X .
- Let’s use the notation $dP(x)$ to refer to the probability dP of the atom x .
- Then, for an event $S \in \Sigma$, $\int_S dP(x)$ is the sum of the probabilities of all the atoms of X in S (and thus is the “probability” of S).
- This means that $\int_S dP(x) = P(S)$

This becomes PDFs and PMFs!

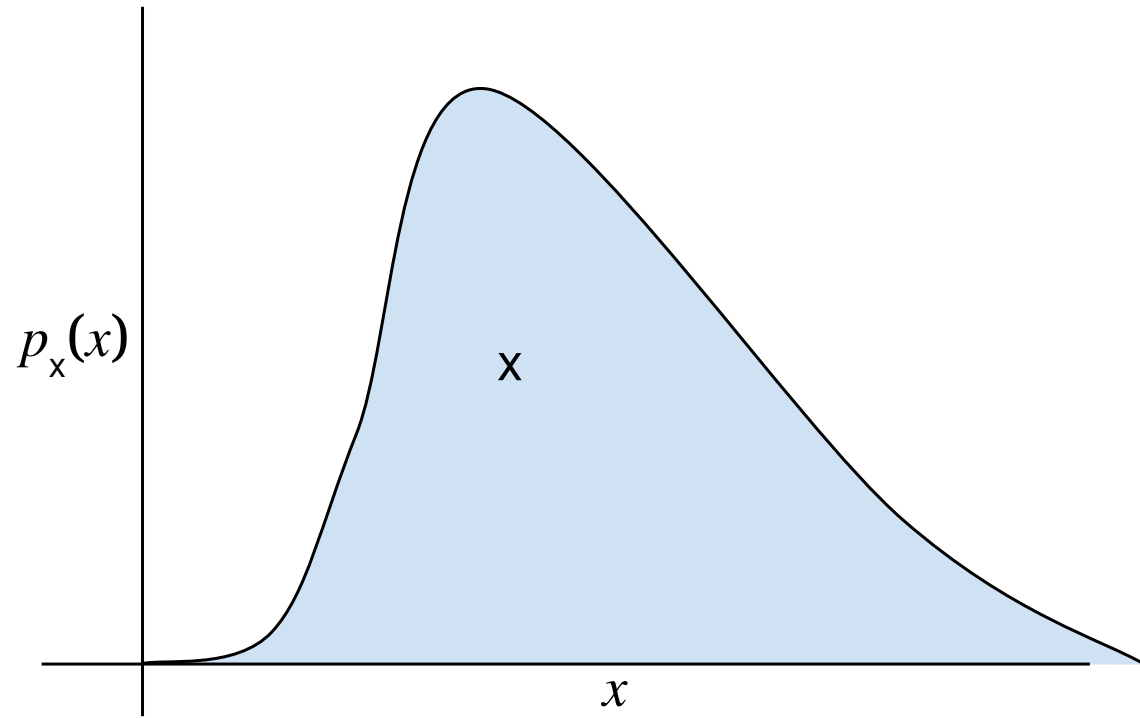
- Suppose x is some “atom” of X .
- Let’s use the notation $dP(x)$ to refer to the probability dP of the atom x .
- Then, for an event $S \in \Sigma$, $\int_S dP(x)$ is the sum of the probabilities of all the atoms of X in S (and thus is the “probability” of S).
- This means that $\int_S dP(x) = P(S)$
- Suppose that each atom of X has size dx . Then I can write:
$$\int_S (dP(x)/dx) dx = P(S)$$

This becomes PDFs and PMFs!

- We will use the notation $p_x(x) = (dP(x)/dx)$ to denote the PDF/PMF.
- For a discrete random variable (X is finite), $p_x(x) = P(\{x\})$.
- For a continuous random variable (X is infinite), $p_x(x) \neq P(\{x\})$ in general.

Putting it all together...

Let x be a random variable defined with PDF/PMF $p_x(x)$, then for any event S , $P(S) = \int_S p_x(x) dx$.



What does $\int_S p_x(x) dx$
really mean?

$$\int_S p_x(x) \, dx = \int_X \mathbb{1}_{x \in S}(x) p_x(x) \, dx$$

$$\begin{aligned}
 \int_S p_x(x) \, dx &= \int_X \mathbb{1}_{x \in S}(x) p_x(x) \, dx \\
 &= \overrightarrow{\mathbb{1}_{x \in S}} \cdot \overrightarrow{p_x}
 \end{aligned}$$

Expectation of a
random variable

Typical “definition”

The expected value of a random variable is the weighted average of all the possible values it can take: $\sum_x x P(\{x\})$

The “Expectation” operator computes the inner product of a function and the probability distribution

Let $f: X \rightarrow \mathbb{R}$

$$\begin{aligned}\mathbb{E}_x[f] &= \int_X f(x) p_x(x) dx \\ &= \vec{f} \cdot \vec{p}_x\end{aligned}$$

Expectation is *the* fundamental operation over random variables.

- $P(S) = \mathbb{E}_x[\mathbb{1}_{x \in S}]$
- $\mu = \mathbb{E}_x[x]$
- Moments of a distribution are expectations of powers.

Properties of Expectation

- Expectation is linear.
- In certain circumstances, we can commute expectation and differentiation.

Joint, conditional, and
marginal random
variables

Suppose x and y are
random variables.

What is the distribution
for $x + y$?

Suppose x and y are
random variables.

What is the distribution
for $(X, \Sigma, P) + (Y, \Sigma', P')$?

$x + y$ can't be defined unless
I have an element-wise way
to relate (X, Σ, P) & $(Y, \Sigma',$
 $P')$!

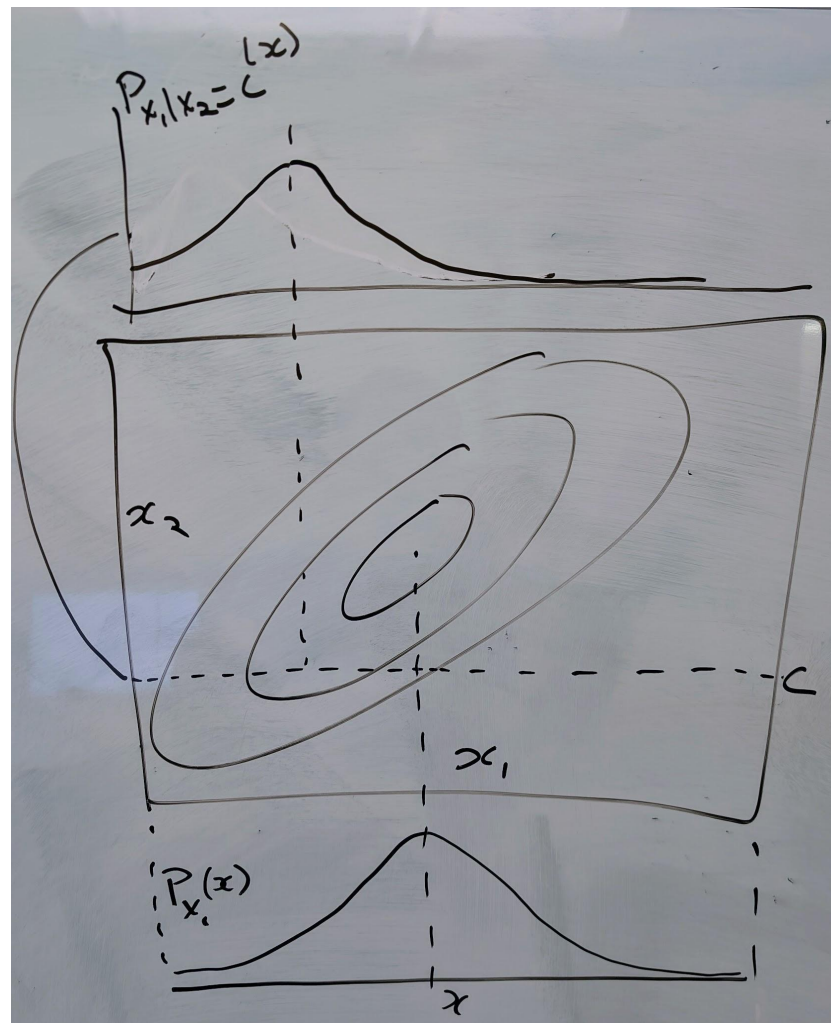
Solution: Define

$$(X \times Y, \Sigma \times \Sigma', P \times P')$$

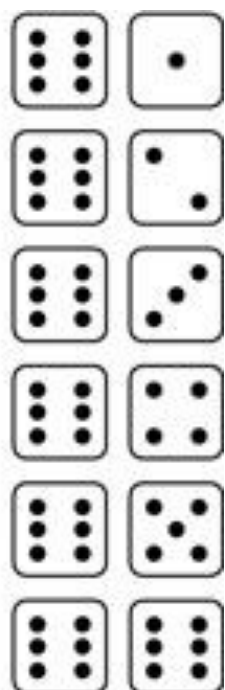
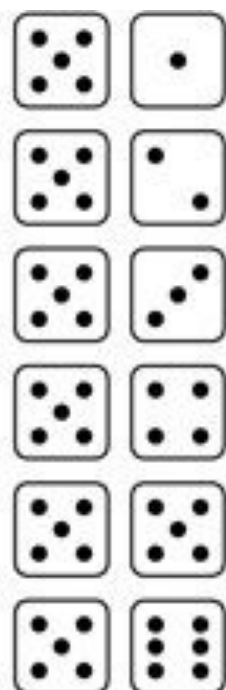
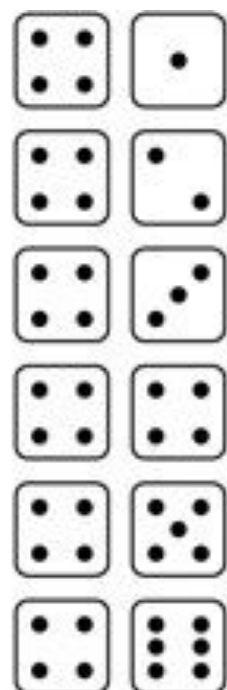
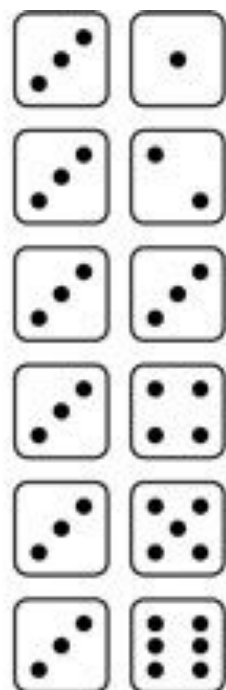
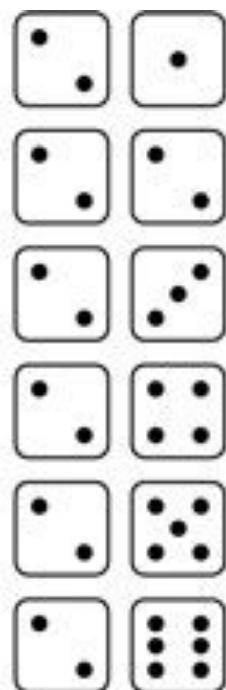
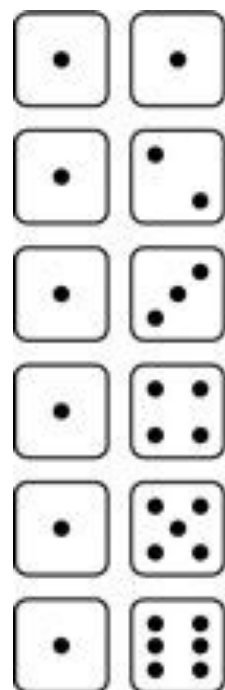
Solution (alternate):

Define Z such that

$Z \rightarrow X$ and $Z \rightarrow Y$



Example: Two dice!



Example: Two dice
and their sum!

